

David Mukuruva

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SUMMARY

Mathematics & Biomedical Engineering undergraduate (Dartmouth '27) working at the intersection of pure mathematics (complex analysis, algebra), scientific computing, and quantitative problem-solving. Research in finite Blaschke products and inscribed ellipses; teaching across introductory physics; builder of tested software and modeling tools.

EDUCATION

Dartmouth College

Hanover, NH

B.A., Double Major in Mathematics and Biomedical Engineering

Expected June 2027

- **Mathematics:** MATH 24 Honors (Linear Algebra, proof-based – Friedberg/Insel/Spence); MATH 113 (Analysis); continued self-study in abstract algebra and PDEs
- **Engineering & Science:** Biotechnology & Biochemical Engineering; Cell Biology; Genetics; Introduction to Biomedical Engineering

Thayer School of Engineering

Hanover, NH

Bachelor & Master of Engineering in Biomedical Engineering

June 2028

RESEARCH EXPERIENCE

Polymath Jr – “Finding Ellipses” (PI: Prof. Y. Zeytuncu)

June 2025 – Present

Research Assistant → Senior Thesis (advisor: Prof. D. Williams)

Remote / Hanover, NH

- Confirmed both target problems were still open before committing a research year to them, by auditing the complete citation graph of the field’s key survey (arXiv:2206.07466) through 2024–2025 results rather than inheriting the prior work’s framing
- Attacking whether the Blaschke curve of $B_3 \circ B_2$ is always an ellipse via degree-6 decomposition and the Kippenhahn/Darboux-curve criterion; separately constructing explicit odd-order ($n = 5$) Poncelet ellipses by extending a cutoff CMV-matrix / orthogonal-polynomial method established only for even orders
- Built a MATLAB tool for 6-dimensional numerical characterization and automated data generation during the Polymath phase; current numerics are in Python/Sage with selected results formalized in Lean
- Work presented (contributed talk) with the Polymath Jr group at the Joint Mathematics Meetings (JMM), 2026; the basis of prospective senior-thesis work

FitzLab, Thayer School of Engineering (Prof. M. Fitzpatrick)

2026 – Present

Undergraduate Researcher – quantum control benchmarking (superconducting-microwave platform)

Hanover, NH

- Showed reinforcement-learned quantum-control outcomes form a mixture distribution rather than a Gaussian, by designing and running a 40-run seed sweep on Dartmouth’s SLURM cluster (40/40 complete, zero failures): 2 of 8 seeds on the hardest target freeze at the do-nothing baseline, verified to 10 significant figures against an independent calculation — a failure mode that mean-and-error-bar reporting hides
- Overturned a “tight, unimodal” group result by removing post-selection: unconditioned on the ancilla, its dispersion doubled and the per-seed ranking reordered, showing the conditioning rather than the control method had produced the effect
- Independently reproduced an internal benchmark’s structure to machine precision but not its headline, traced the gap to parameters the source note never states, and labeled the claim not-reproducible-as-documented; separately traced the sweep’s failure to converge within budget to PPO gradient-estimate variance, showing that a larger batch with smaller action-noise converges where the shipped defaults are too slow
- Built and maintain the open-source SLURM seed-sweep toolkit the group runs on (github.com/dmukuruva-creator/quantum_control_rl_server); authored a consolidated literature review (~46 pp, ~80 sources) mapping the lab’s superconducting-microwave program from foundations to current results

Thayer School of Engineering – Goods Lab

Jan 2024 – Sep 2024

Research Assistant (Computational)

Hanover, NH

- Analyzed single-cell RNA data from gynecological-cancer patients in R under faculty supervision, building reproducible pipelines
- Characterized diseased-cell signalling pathways using Seurat, CellChat, and Bioconductor
- Cross-referenced 16 cell identities with Enrichr and patient metadata to surface candidate biomarkers and associations

TEACHING

Dartmouth Physics Department

Apr 2025 – Present

Teaching Assistant & Grader

Hanover, NH

- Facilitate 10 weekly labs across 4 introductory Physics courses, guiding 40+ students through experimental procedures
- Grade prelab exercises and return instructional feedback within 48 hours
- Field conceptual and procedural questions during labs and calibrate grading with faculty to keep feedback consistent across sections

Dartmouth Emerging Engineers (DEE)

Apr 2025 – Present

Lead Physics Tutor

Hanover, NH

- Designed the physics support curriculum and problem sets for the DEE program, aligned to the introductory physics sequence
- Lead weekly office hours (8–12 students per session), reinforcing core concepts from the introductory sequence
- Mentor and coordinate peer tutors, tracking student progress to target common misconceptions

Certification: Learning Fellows Program – Pedagogy Series, DCAL (May 14, 2026).

Mentoring: Directed Reading Program – Combinatorial Game Theory (mentor: M. Polley), Spring 2026.

SELECTED PROJECTS

QuBlitz | *Python, JavaScript*

2025 – Present

- Designed an interactive quantum-mechanics learning game (built within the Fitzpatrick Lab) that builds student intuition for phase interference and Bell-pair entanglement through hands-on play
- Structured levels around core quantum concepts, turning abstract formalism into intuitive, playable challenges for introductory learners
- Implemented and unit-tested the underlying physics engine with a passing automated test suite

Studium | *Python, JavaScript*

2025 – Present

- Built a local-first AI study app (Python standard-library server + single-page JS front end) that turns uploaded PDFs into quizzes, flashcards, and concept maps
- Integrated a multi-provider AI proxy (Anthropic, OpenAI, Google) with API keys in macOS Keychain and user data kept in-browser (IndexedDB) for a fully local, private workflow
- Added a quantitative Quant-Drill module and hardened the server (SSRF guards, request-body caps) with Node and Python test suites

Aleph-Nought | *Next.js (TypeScript), FastAPI, PostgreSQL*

2025 – Present

- Built a full-stack interactive STEM learning platform covering six Dartmouth math/physics course syllabi (MATH 1/3/8/13, PHYS 13/14); deployed to production on Vercel/Render
- Built a symbolic exercise engine that parses and evaluates student-entered mathematical expressions for automated exercise grading
- Engineered an automated content pipeline – Manim animations, FFmpeg video processing, Whisper transcription, and interactive Plotly figures – with Redis caching and Cloudflare R2 asset storage
- Integrated AI-generated feedback (Groq, Llama-3.3-70B) and designed a three-tier open/paid/enrichment content and licensing model

PRESENTATIONS

“Finding Ellipses: Blaschke Products, Poncelet’s Theorem, and the Numerical Range,” Joint Mathematics Meetings (JMM), 2026 (Polymath Jr research group).

TECHNICAL SKILLS

Programming: Python (NumPy/SciPy), Sage, C, MATLAB, R, Lean; Git, SLURM/HPC

Methods: numerical methods, linear algebra, probability, proof-based mathematics, stochastic optimization, reproducibility & experiment design, data analysis

Engineering/CAD: SolidWorks, TinkerCAD, Arduino IDE

LEADERSHIP & ACTIVITIES

Undergraduate Advisor (Residential Advisor); Archery; Chess.

References: available upon request.